

Logic Quiz Test Report

Test Date: February 25, 2026

Test Subject: Butler AI Agent

Evaluator: Zheng Weisi (AI Assistant)

Overall Scoring

Dimension	Score	Notes
Accuracy	92/100	Mostly correct, minor deviations in some answers
Reasoning Depth	95/100	Detailed reasoning using multiple methods
Clarity	90/100	Clear structure, some answers could be more concise
Innovation	85/100	Standard solutions, lacking unique perspectives
Total	90.5/100	Excellent

Question 1: Prisoner's Dilemma Variant [92/100]

Three prisoners choose to confess or not. Nash equilibrium: All confess, 5 years each.

Strengths: Correct payoff matrix, dominant strategy identification.

Improvement: Could discuss repeated game cooperation.

Question 2: Pirate Coin Division [98/100]

5 pirates divide 100 coins. Captain's optimal: 98, 0, 1, 0, 1

Strengths: Perfect backward induction, correct vote counting.

Excellent demonstration of game theory principles.

Question 3: Monty Hall Problem [95/100]

Switch doors: $\frac{2}{3}$ win probability. Stay: $\frac{1}{3}$ probability.

Strengths: Correct probability calculation, intuitive explanation.

Could add Bayesian theorem proof for rigor.

Question 4: Liar Paradox [90/100]

"All my statements are false" - self-referential paradox.

Solutions: Language hierarchy, context restriction, multi-valued logic.

Good connection to Godel's incompleteness theorem.

Question 5: Blue-Eyed Islanders [99/100]

100 blue-eyed islanders commit suicide on day 100.

Strengths: Perfect induction, common knowledge concept.

Excellent distinction between 'at least 1' vs 'at least 2'.

Question 6: Einstein's Logic Puzzle [98/100]

Who keeps fish? German. Who drinks water? Norwegian.

Strengths: Complete step-by-step deduction, correct answer.

Well-organized elimination process.

Question 7: Hilbert's Hotel [97/100]

Infinity + 1 = infinity, infinity + infinity = infinity.

Strengths: All three scenarios solved correctly.

Clear explanation of countable infinity properties.

Question 8: Unexpected Hanging [91/100]

Thursday execution IS unexpected. Prisoner's backward induction fails.

Core insight: Confusing post-hoc certainty with pre-hoc uncertainty.

Could explore modal logic perspectives further.

Question 9: Newcomb's Paradox [95/100]

Two-box vs one-box decision. Recommended: Take only Box B (~99M expected).

Strengths: Correct EDT vs CDT comparison, accurate calculations.

Good coverage of decision theory■■■.

Question 10: Barber Paradox [95/100]

Barber who shaves all who don't shave themselves - contradiction.

Solutions: ZFC axioms, type theory, Tarski's truth theory.

Strong connection to Russell's paradox.

Summary and Conclusion

Score Summary

Question	Score	Grade
Question 1	92/100	Excellent
Question 2	98/100	Outstanding
Question 3	95/100	Outstanding
Question 4	90/100	Excellent
Question 5	99/100	Outstanding
Question 6	98/100	Outstanding
Question 7	97/100	Outstanding
Question 8	91/100	Excellent
Question 9	95/100	Outstanding
Question 10	95/100	Outstanding
Average	90.5/100	Excellent

Overall Assessment

The Butler AI Agent performed excellently in this high-difficulty logic quiz test, achieving a total score of 90.5/100.

Strengths:

- Excellent logical reasoning ability, all answers fundamentally correct
- Skilled in using induction, backward reasoning, and other methods
- Clear expression and complete structure

Areas for Improvement:

- Some questions could introduce more diverse analytical perspectives
- Philosophical questions could explore conceptual essence more deeply
- Could attempt to propose original insights rather than standard solutions

Final Grade: A (Excellent)